

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Power Electronics	Course Code:	EE3032
	Program:	BS Electrical Engineering	Semester:	Spring 2025
	Assignment Submission Date:	27-February 2025	Total Marks:	50
	Section:		Weight:	4
	Exam Type:	BEE-6A	Page(s):	1
		Assignment- 3 (Problem Based Learning)	CLO #	3

Student Name: _____ **Roll No.** _____ **Section:** _____

- Instructions:**
1. Use A4-sized sheets for this assignment.
 2. Avoid cutting and overwriting.
 3. Secure the pages with staple, clip, or binding. Do not submit loose pages.
 4. Assignment must be submitted on or before deadline at google classroom.
 5. Late submission will incur a penalty of 20% per day. After 2 days, assignment will no longer be accepted.

Problem Statement:

In high-speed power electronics applications, MOSFETs are widely used for switching operations due to their efficiency and fast response. However, the choice of the gate driver significantly impacts the performance of the circuit. A 7406 open-collector inverter is often considered for MOSFET switching, but it has several limitations in driving high-speed MOSFETs, especially in high-side configurations.

Question No. 1 (CLO No. 3) **Marks: 15**

What might happen if you design a high side MOSFET switch with 7406 drive to work at more than 30 kHz? Can you elaborate on the reason?

Question No. 2 (CLO No. 3) **Marks: 20**

Design an improved gate drive circuit or propose an alternate solution to solve the problem highlighted in Question 1.

Question No. 3 (CLO No. 3) **Marks: 15**

Justify your proposed solution while comparing it with 7406 drive in terms of switching speed, power dissipation and efficiency.

ASSIGNMENT #023

Q.1

Using a 7406 open-collector inverter to drive a high-side MOSFET switch at frequencies above 30kHz presents several problems:-

1) Switching Speed Limitations

- ⇒ The 7406 inverter has significant propagation delay and slow rise/fall times, which become problematic at higher switching frequencies.
- ⇒ The pull-up resistor required for the open-collector output further increases the RC time constant, slowing down the turn-on transition of the MOSFET.
- ⇒ At high frequencies, the MOSFET may not fully turn ON/OFF within each cycle, leading to excessive power loss.

2) Limited Current Drive Capability

High speed MOSFETs typically have significant gate charge requirements. The 7406 may not be able to supply the necessary current to charge and discharge the gate capacitance quickly enough, leading to slower turn-on and turn-off times.

This can result in increased switching losses and reduced overall efficiency.

3) Voltage Spikes and EMI Issues

- ⇒ The slow switching transitions can cause ringing and voltage overshoots, introducing electromagnetic interference (EMI) issues.
- ⇒ These voltage spikes can damage other components in the circuit.

4) Voltage Level Mismatch

The 7406 operates at TTL logic levels (i.e. 0V-5V), while high-speed MOSFETs often require gate voltages of 10V-15V for optimal switching.

5) Increased Power Dissipation and Heating

- ⇒ Due to slow switching behavior, the MOSFET will spend more time in the linear region, leading to increased conduction losses and switching losses.
- ⇒ The system will suffer from higher temperature rise, reducing efficiency and increasing the risk of MOSFET failure.

Q.1 No.2

To overcome the limitations of the 7406 inverter, we can use a dedicated gate driver IC like IR2110, specifically designed to drive MOSFETs efficiently.

Advantages of Gate Driver IC

- ⇒ Can source/sink several amperes of current, ensuring fast switching.
- ⇒ Can provide gate voltages up to 15V, suitable for power MOSFETs.
- ⇒ Uses an internal bootstrap capacitor to provide the necessary voltage for driving high-side mosfets.
- ⇒ Capable of driving high-side and lowside mosfet efficiently.
- ⇒ By reducing switching delays and losses, the overall efficiency of circuit is improved.

Proposed Gate Driver Circuit

- ⇒ Use an IR2110 half-bridge gate driver
- ⇒ Connect the low-side mosfet gate directly to the driver output.
- ⇒ Use a bootstrap capacitor (10 μ F) and diode to provide the necessary high-side gate drive.
- ⇒ Supply the IR2110 with 12V to 15V for optimal voltage.
- ⇒ Use a resistor (10 Ω) in series with the gate to control the turn-on and turn-off speed.

Q. No. 3

7406 Open-Collector
Drive

IR2110 Gate Driver

Parameters

Switching Speed

Slow (high propagation delay)

Fast (low propagation delay)

Current Drive

Low ($< 20\text{mA}$)

High (2A-3A)

Voltage Drive

Limited to 5V (TTL Logic).

Up to 15V (suitable for mosfets)

Power Dissipation

High (due to slow switching, mosfet stays longer in linear region)

Low (due to fast switching)

Efficiency

Low (more losses)
Poor at high frequencies)

High (reduced losses)

High-Frequency Operation

Not suitable beyond 30kHz

Can operate above 100kHz easily

Heat Generation

High (due to slow switching)

Low (due to efficient transitions)

High-Side Drive

Requires extra circuitry

Built-in bootstrap circuit.

1) Higher Switching Speed

- ⇒ The IR2110 gate driver enables fast rise and fall times, allowing the mosfet to switch ON/OFF efficiently.
- ⇒ This minimizes the time spent in the linear region, reducing switching losses.

2) Lower Power Dissipation

- ⇒ The 7406 drive causes excessive heat generation, leading to efficiency loss.
- ⇒ A gate driver IC ensures the mosfet is fully on or off, reducing heat dissipation.

3) Better Efficiency

- ⇒ Since the mosfet turns ON/OFF quickly, the system operates at higher efficiency with minimal energy loss.
- ⇒ This is critical for power electronics applications, such as motor drivers, inverters and SMPS.